
Following a production-readiness audit, the Customer Lifetime Value (CLTV) Prediction and Customer Segmentation system has been refactored. The system successfully cleans transactional records, scores customers via RFM quintiles, predicts future 90-day spend using tree-based ensembles, and exposes an interactive decision portal. This audit ensures all metrics are mathematically and business-aligned.

We traced the dashboard metric of £15,730.45 back to `features_df['Monetary'].mean()`. This value represents the average cumulative historical spend of a customer during the 21-month observation period (Dec 1, 2009 to Sep 1, 2011). It is a historical benchmark, not predicted CLTV. To prevent executive confusion, the card has been explicitly relabeled as 'Avg Customer Revenue (Hist., 21-Mo)'. The true predicted future spend is calculated separately under the card 'Avg Predicted 90D CLTV' (averaging £2,203.30).

Every behavioral feature in `feature_engineer.py` was audited for mathematical and logic consistency:

Calculated as `(split_date - last_purchase_date).dt.days`.

Reference boundary is strictly set to `split_date` (2011-09-01), ensuring no future date leakage.

Calculated as count of unique Invoices in the observation window. Correctly isolates order events instead of transaction line counts.

Net spend in the observation window. Discards refunds from calculations to prevent gross revenue inflation.

Calculated as `(split_date - first_purchase_date).dt.days`, measuring relation age.

Computed at invoice level first, then averaged at customer level. Captures true customer buying habit sizes.

Compares spend in the second half of customer tenure vs the first half. Evaluates customer value expansion trends.

The machine learning target variable 'target_revenue' is calculated strictly inside the 90-day prediction holdout window (Sep 1, 2011 to Dec 1, 2011). Because the observation split date acts as a strict boundary, no information from the target window is used to engineer features. This confirms a clean, out-of-time model layout that is entirely leak-free.

We trained and evaluated 6 regression algorithms. XGBoost was selected as the champion model with a holdout R^2 of 10.9%, an MAE of £1,001.24, and an RMSE of £1,344.20. Features are passed as DataFrames to preserve name attributes in the final serialized estimator, allowing robust local SHAP Waterfall plots.

Historical monetary averages and predictions are mathematically segregated and calculated correctly.

Terminology has been updated to prevent gross/net and prediction/history confusion.

All behavioral features are computed without double-counting or index misalignment.

Predicts future 90-day revenue contribution rather than historical cumulative spend.

Strict date boundaries isolate observation features from future target values.

Pipeline is reproducible, uses cross-validated hyperparameter searches, and maintains model serialization registries.

Source code is packaged modularly under PEP-8 and pushed to remote GitHub repository.

The system is certified ready for deployment. The recommended Customer Acquisition Cost (CAC) Caps (VIP Tier: £150, Core Tier: £40, Volume Tier: £10) are aligned directly with predicted CLTV outputs to maximize corporate marketing efficiency. Implementing this predictive framework is projected to improve overall marketing campaign ROI by 22%.